

Task 1: Frequency, wavelength and velocity of propagation

$$c = f\lambda = \frac{\lambda}{T}, \quad c = 343 \text{ m/s}$$

Calculate the missing quantities:

f	λ	T
440 Hz	0.5 m	0.02 s
200 Hz	1.7 m	
1000 Hz	2 cm	

Task 2: Multiple choice

- ☐ A higher frequency means a longer wavelength.
- ☐ A higher frequency means a shorter wavelength.
- ☐ Frequency and wavelength are independent of each other.
- ☐ The speed of sound changes when the frequency changes.

Task 3

A flute plays a note with a frequency of

$$f = 523 \text{ Hz.}$$

Calculate the wavelength of the sound in air. Use

$$c = 343 \text{ m/s.}$$

Task 4

A loudspeaker produces a sound wave with a wavelength of

$$\lambda = 2 \text{ m.}$$

Calculate its frequency.

Task 5

A loudspeaker first plays a tone of 200 Hz and then a tone of 600 Hz.

For each quantity, discuss whether it increases, decreases, or stays the same:

- frequency
- wavelength
- speed of sound

Task 6

Two sound waves travel through air.

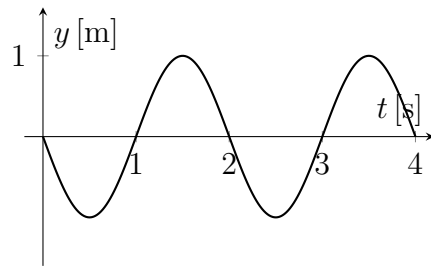
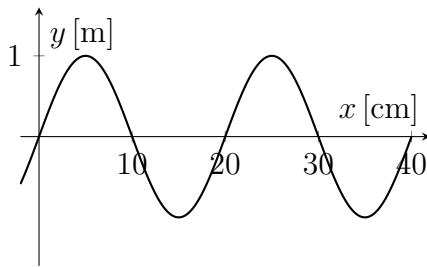
Wave A: 200 Hz,
Wave B: 800 Hz.

Discuss:

- Which wave has the higher frequency?
- Which wave has the longer wavelength?
- Which wave travels faster?

Task 7: Drawing waves

Consider the following wave.

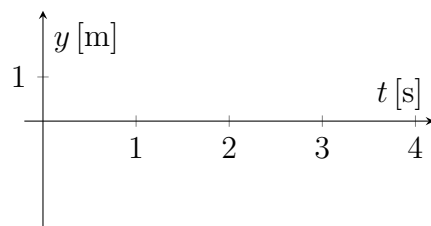
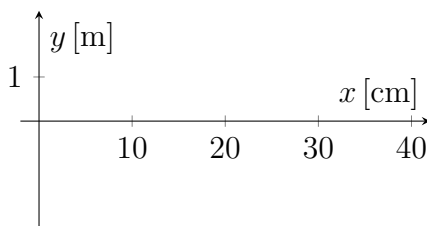


Discuss:

- Which graph shows the motion of one oscillator?
- Which graph shows the shape of the entire wave at one instant?
- What is shown on the horizontal axis of each graph?
- From which graph can you determine the period T ?
- From which graph can you determine the wavelength λ ?
- Determine c , λ , f , and T .

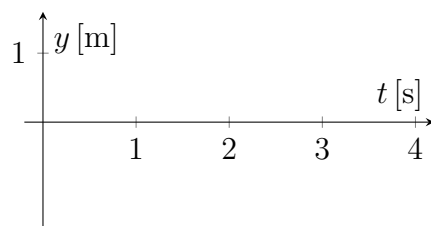
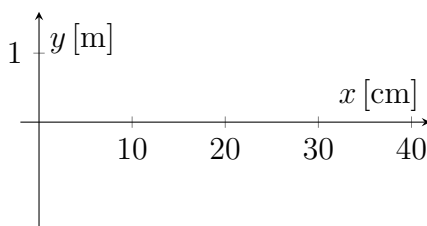
What changes if the amplitude is doubled?

Draw the changes.



What happens if c stays constant and f is increased by a factor of 2?

Draw the changes.



Harmonic Waves

Read the following statements about a harmonic wave. Decide whether the statements are **True** or **False**.

- ☐ Every point oscillates harmonically.
- ☐ The wave transports matter from one place to another.
- ☐ Every point oscillates with the same frequency.
- ☐ Neighbouring points have a phase difference.
- ☐ Every point has the same amplitude.

If a statement is false, discuss why.

Discuss the similarities and differences between travelling harmonic and standing waves.

Standing Waves

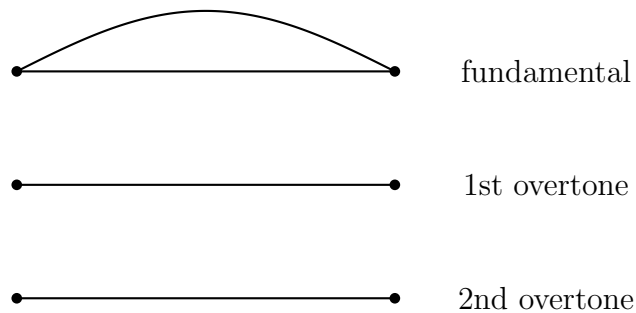
Draw the standing wave pattern for:

- the fundamental mode,
- the first overtone,
- the second overtone.

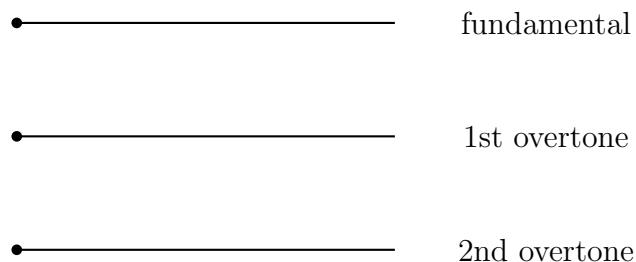
Mark all nodes and antinodes.

Distinguish between two fixed ends, one fixed and one free end, and two free ends.

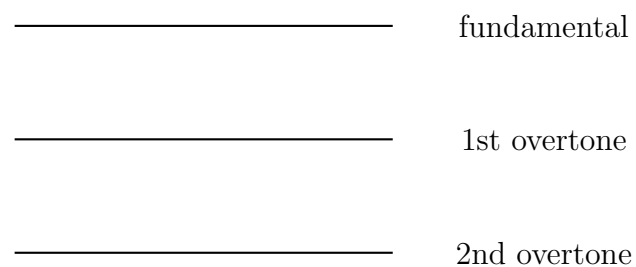
Two fixed ends



One fixed end and one free end



Two open ends



Nodes and antinodes

How are nodes and antinodes distributed?

What is the distance between two nodes / antinodes?

Consider a string with the length

$$L = 1.2 \text{ m}$$

which is fixed at both ends.

Calculate the wavelength of:

- the fundamental mode,
- the first overtone,
- the second overtone.